Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (EE) Examination

March 2018 (Backlog)

EE401/EE401.01 Electrical Machine Design I

Date: 26.03.2018, Monday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Explain in detail the design of armature in context with following points. (a) Number of armature conductors (b) Number of armature coils (c) Number of armature slots. **[07]**

Q – 2.a Enlist the methods for reducing the effect of armature reaction. Also explain the effect of increasing length of airgap at pole tips on armature reaction. **[07]**

Q – 2.b What are the points to be considered for selection of D and L of DC machine? **[07]**

OR

Q – 2.a How peripheral speed and moment of inertia affects the main dimensions of DC machine? Explain in detail. **[07]**

Q – 2.b With necessary diagram, explain how commutating poles can reduce the effect of armature reaction? **[07]**

Q – 3 **Answer Any Two. (Each question carry seven marks)** **[14]**

(1) Derive output equation of DC machine.

(2) Discuss the points to be taken into consideration for selection of specific electric loading in rotating machines.

(3) A 275kW, 550V, 900rpm, 6 poles, wave wound DC machine has 180 slots with 8 conductors in each slot and 70% of armature surface covered by poles.

(a) Find armature mmf per pole at rated armature current

(b) Find number of conductors that should be placed in each pole face to provide adequate compensating winding

(c) Estimate the number of turns required on each of 6 commutating pole to provide a flux density across an effective gap length of 5mm if machine has no compensating winding.

SECTION – II

Q - 4 Write down the detail design procedure for DC shunt field winding. **[07]**

Q – 5.a State and explain the factors affecting on design of the machine? **[07]**

Q – 5.b Derive the output equation for single phase transformer. **[07]**

OR

Q – 5.a If a designer selects higher value of flux densities what will be its effect on performance and cost of the transformer? Explain in detail. **[07]**

Q – 5.b How Faraday's law of electromagnetic induction is applied in design of an electrical machine? Explain in detail. **[07]**

Q – 6. **Answer Any Two. (Each question carry seven marks)** **[14]**

(1) Write down the detailed design procedure for HV and LV winding of transformer.

(2) Derive an output equation of three phase transformer.

(3) Derive an equation for optimum design of transformer for minimum cost and minimum losses?

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